

LAB 6

Analog Interface ADC

Objectives:

- Understand and implement ADC in embedded systems.
- Interface a potentiometer to an ADC input.
- Display ADC readings on a terminal or an LCD.
- Explore the impact of different analog inputs on ADC performance.

Material:

- Keil IDE
- TI Tiva development board
- A Terminal software like Tera Term or Putty.
- 1602 LCD (optional)
- 10K POT

Submission Instructions:

- Demonstrate your work for each activity to your instructor. Missing the instructor signature results in 20% penalty.
- Submit a zipped file of your project folder onto the BB dropbox. Missing the zipped project file results in a 20% penalty.
- Write a lab report detailing the implementation and results. Include the schematic and picture of your circuit, screenshot of your code and results. Answer all the questions.

Lab Activity 1 [30 points]

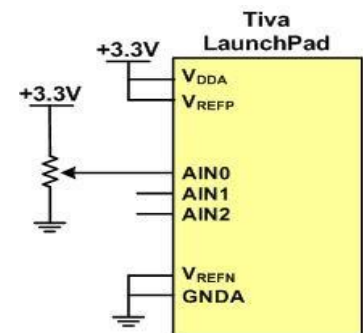
Objective:

Use ADC to read an analog input from a potentiometer and display the results via UART.

Steps:

- Connect the 10K POT to analog input AIN0.
- Write a program to read the analog input and send the value to the terminal via UART.
- Open Tera Term or Putty and configure it to display the ADC values.

- Change the potentiometer position and observe how the output value changes in the terminal.



- Adjust the potentiometer to a random position and observe the ADC reading. Theoretically calculate the corresponding analog voltage using the ADC formula. Then, measure the actual voltage using a voltmeter. Compare the calculated and measured values, explaining any discrepancies. Show all calculation steps and analysis.

Lab Activity 2 [40 points]

Objective:

Modify the circuit and program from Lab Activity 1 to use a different ADC input.

Steps:

1. Connect the **potentiometer to AIN4** instead of AIN0.
2. Update your code to read from **AIN4**.
3. Transmit the ADC readings via **UART** and display them in **Tera Term or Putty**.
4. Compare the ADC values from **AIN0 and AIN4**.

Questions [30 points]

1. What is the significance of ADC resolution in this exercise?
2. How does reference voltage affect ADC readings?
3. What are some possible sources of error in ADC readings?
4. Explain the process of converting an analog signal into a digital value.
5. What are the possible values of the ADC output for a **3.3V reference voltage**? What about a **5V reference voltage**?
6. A sensor outputs a voltage ranging from **0V to 2.5V** with a sensitivity of 1mV (0.001V) per unit change.
 - Can the TIVA C ADC accurately capture the sensor's smallest change in voltage?
 - Justify your answer by calculating the ADC's voltage resolution per step.
 - Suggest two ways to improve the accuracy of temperature measurements. Assume that you can change the hardware if necessary.